

This assignment used logistic regression in R to evaluate customer churn in a telecommunications dataset. A full model was first created using all available variables, followed by additional models using the three most significant predictors compare how each performed individually and collectively in predicting customer churn.

Figure 1 shows the initial preparation steps before building the logistic regression models. After loading the dataset into R, the `str()` function was used to review the structure of each variable and confirm that the dataset loaded correctly. Most of the predictor variables were already structured as factors, which was important because logistic regression in this assignment relied heavily on categorical inputs. Since these variables were already stored as factors, no additional recoding was needed before modeling. The output also confirmed that the dataset contained 7,032 observations across 21 variables, after which the data was split into training and testing partitions so model performance could be evaluated on unseen data.

Figure 1

Initial Data Load, Structure Review, and Training/Test Partition Setup

```
> #####
> #Critical Thinking 5
> #####
> ##Read CSV file & view Data Structure
> getwd()
[1] "c:/Users/schwe/OneDrive/Documents/School/Data_Science_Foundations/data"
> telco <- read.csv("MIS470Telcocustomer.csv", stringsAsFactors = TRUE)
> str(telco)
'data.frame': 7032 obs. of 21 variables:
 $ x      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ gender : Factor w/ 2 levels "Female","Male": 1 2 2 2 1 1 2 1 1 2 ...
 $ SeniorCitizen : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 1 1 1 1 ...
 $ Partner      : Factor w/ 2 levels "No","Yes": 2 1 1 1 1 1 1 1 2 1 ...
 $ Dependents   : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 2 1 1 2 ...
 $ PhoneService : Factor w/ 2 levels "No","Yes": 1 2 2 1 2 2 2 1 2 2 ...
 $ MultipleLines : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 2 2 1 2 1 ...
 $ InternetService : Factor w/ 3 levels "DSL","Fiber optic",...: 1 1 1 1 2 2 2 1 2 1 ...
 $ OnlineSecurity : Factor w/ 2 levels "No","Yes": 1 2 2 2 1 1 1 2 1 2 ...
 $ OnlineBackup   : Factor w/ 2 levels "No","Yes": 2 1 2 1 1 1 2 1 1 2 ...
 $ DeviceProtection : Factor w/ 2 levels "No","Yes": 1 2 1 2 1 2 1 1 2 1 ...
 $ TechSupport    : Factor w/ 2 levels "No","Yes": 1 1 1 2 1 1 1 1 2 1 ...
 $ StreamingTV    : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 2 2 1 2 1 ...
 $ StreamingMovies : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 2 1 1 2 1 ...
 $ Contract       : Factor w/ 3 levels "Month-to-month",...: 1 2 1 2 1 1 1 1 1 2 ...
 $ PaperlessBilling : Factor w/ 2 levels "No","Yes": 2 1 2 1 2 2 2 1 2 1 ...
 $ PaymentMethod  : Factor w/ 4 levels "Bank transfer (automatic)",...: 3 4 4 1 3 3 2 4 3 1 ...
 $ MonthlyCharges : num  29.9 57 53.9 42.3 70.7 ...
 $ TotalCharges   : num  29.9 1889.5 108.2 1840.8 151.7 ...
 $ churn          : Factor w/ 2 levels "No","Yes": 1 1 2 1 2 2 1 1 2 1 ...
 $ tenure_group   : Factor w/ 5 levels "> 60 Month","0-12 Month",...: 2 4 2 4 2 2 3 2 4 1 ...
> # install.packages("caret")
> library(caret)
> #####
> # Prep data. Partition data into training and testing partitions
> intrain <- createDataPartition(telco$churn, p = 0.7, list = FALSE)
> set.seed(2017)
> training <- telco[intrain, ]
> testing <- telco[-intrain, ]
```

Note. The dataset was loaded into R as a data frame and `str()` output was used to confirm variable types before modeling. The data was then partitioned into 70% training data and 30% testing data to prepare for logistic regression analysis.

Figure 2 shows the full logistic regression model after it was evaluated against the testing dataset using a 0.5 probability threshold to classify whether a customer was predicted to churn or remain. The confusion matrix shows that the model correctly identified 1,414 non-churn cases and 270 churn cases, while misclassifying 134 non-churn cases and 290 churn cases. Using the misclassification error, the model produced an accuracy of 79.9%, meaning it correctly classified most customer outcomes overall. I also tested alternative classification thresholds (0.3 and 0.8) to see whether model sensitivity would improve; however, both reduced overall accuracy and created less balanced prediction results, so the original 0.5 threshold was retained.

Figure 2

Full Logistic Regression Model Output, Confusion Matrix, and Accuracy for Customer Churn

```
> #####
> # Create a Logistic model for churn with training data
> LogModel <- glm(Churn ~ ., family = binomial(link = "logit"), data = training)
> summary(LogModel)

call:
glm(formula = Churn ~ ., family = binomial(link = "logit"), data = training)

coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -2.582e-01  1.011e+00  -0.255  0.798516
X              2.354e-05  1.918e-05   1.228  0.219586
genderMale    -6.116e-02  7.759e-02  -0.788  0.430562
Seniorcitizen 2.216e-01  1.007e-01   2.200  0.027797 *
PartnerYes    1.540e-02  9.234e-02   0.167  0.867522
DependentsYes -1.350e-01  1.063e-01  -1.270  0.204118
PhoneServiceYes 8.292e-01  7.705e-01   1.076  0.281851
MultipleLinesYes 6.283e-01  2.118e-01   2.966  0.003014 **
InternetServiceFiber optic 2.526e+00  9.463e-01   2.669  0.007598 **
InternetServiceNo -2.390e+00  9.595e-01  -2.491  0.012732 *
OnlineSecurityYes 3.117e-02  2.129e-01   0.146  0.883597
OnlineBackupYes 1.876e-01  2.086e-01   0.899  0.368495
DeviceProtectionYes 3.547e-01  2.084e-01   1.702  0.088776 .
TechSupportYes 9.943e-02  2.143e-01   0.464  0.642669
StreamingTVYes 8.476e-01  3.873e-01   2.189  0.028628 *
StreamingMoviesYes 8.342e-01  3.876e-01   2.152  0.031368 *
Contractone year -7.506e-01  1.284e-01  -5.847  5.00e-09 ***
Contracttwo year -1.681e+00  2.137e-01  -7.866  3.66e-15 ***
PaperlessBillingYes 3.777e-01  8.897e-02   4.245  2.18e-05 ***
PaymentMethodCredit card (automatic) -3.707e-02  1.355e-01  -0.274  0.784350
PaymentMethodElectronic check 3.441e-01  1.125e-01   3.058  0.002225 **
PaymentMethodMailed check 2.993e-02  1.364e-01   0.219  0.826311
MonthlyCharges -6.085e-02  3.766e-02  -1.616  0.106188
TotalCharges -8.672e-05  6.770e-05  -1.281  0.200164
tenure_group0-12 Month 1.273e+00  3.767e-01   3.379  0.000727 ***
tenure_group12-24 Month 4.375e-01  3.401e-01   1.287  0.198252
tenure_group24-48 Month 1.360e-01  2.608e-01   0.522  0.601998
tenure_group48-60 Month -7.918e-02  2.149e-01  -0.368  0.712608
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5702.8  on 4923  degrees of freedom
Residual deviance: 4106.5  on 4896  degrees of freedom
AIC: 4162.5

Number of Fisher Scoring iterations: 6

> # Evaluate model with testing data
> testing$Churn <- as.character(testing$Churn)
> testing$Churn[testing$Churn == "No"] <- "0"
> testing$Churn[testing$Churn == "Yes"] <- "1"
> fitted.results <- predict(LogModel, newdata = testing, type = "response")
> fitted.results <- ifelse(fitted.results > 0.5, 1, 0)
> misClasificError <- mean(fitted.results != testing$Churn)
> # Display a confusion matrix using the table() function
> confusionmatrix <- table(Predicted = fitted.results, Actual = testing$Churn)
> confusionmatrix
      Actual
Predicted 0    1
0      1414  290
1       134  270
> # Accuracy
> accuracy <- 1 - misClasificError
> accuracy
[1] 0.7988615
```

Note. Logistic regression output for the full churn model, including coefficient estimates, confusion matrix, and model accuracy.

Using the coefficient output from the full model in Figure 2, Contract, PaperlessBilling, and tenure_group were selected as the three most significant predictors because they showed the strongest statistical significance and the clearest relationship with customer churn (***). Contract contained two highly significant levels, both with negative coefficients, indicating that customers with one-year or two-year contracts were less likely to churn compared with month-to-month customers. PaperlessBilling also showed strong significance, with customers using paperless billing having a greater likelihood of churn. tenure_group was selected because the shortest tenure category had one of the strongest positive relationships with churn, showing that newer customers were more likely to leave. Although each predictor remained statistically meaningful when modeled individually in Figures 3 through 5, all three produced the same confusion matrix and identical accuracy of 73.4%. This occurred because each single-predictor model classified every customer into the majority non-churn category at the 0.5 threshold, which maintained overall accuracy but did not independently improve identification of churn cases.

Figure 3

Logistic Regression Model Using Contract

```
> #####
> # Create and evaluate logistic model using Contract
> LogModelC <- glm(Churn ~ Contract, family = binomial(link = "logit"), data = training)
> summary(LogModelC)

Call:
glm(formula = Churn ~ Contract, family = binomial(link = "logit"),
    data = training)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.30860    0.03868  -7.977  1.5e-15 ***
ContractOne year -1.75426    0.10669 -16.442  < 2e-16 ***
ContractTwo year -3.14666    0.17364 -18.122  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5702.8  on 4923  degrees of freedom
Residual deviance: 4764.0  on 4921  degrees of freedom
AIC: 4770

Number of Fisher Scoring iterations: 6

> # Evaluate model with testing data
> fitted.resultsC <- predict(LogModelC, newdata = testing, type = "response")
> fitted.resultsC <- ifelse(fitted.resultsC > 0.5, 1, 0)
> misClasificErrorC <- mean(fitted.resultsC != testing$Churn)
> # Display a confusion matrix using the table() function
> confusionmatrixC <- table(Predicted = fitted.resultsC, Actual = testing$Churn)
> confusionmatrixC
      Actual
Predicted 0    1
      0 1548  560
> # Accuracy
> accuracyC <- 1 - misClasificErrorC
> accuracyC
[1] 0.7343454
```


Note. Logistic regression output for the Contract-only model, including coefficient estimates, confusion matrix, and model accuracy.

Figure 4

Logistic Regression Model Using PaperlessBilling

```
> #####
> # Create and evaluate logistic model using PaperlessBilling
> LogModelP <- glm(Churn ~ PaperlessBilling, family = binomial(link = "logit"), data = training)
> summary(LogModelP)

call:
glm(formula = Churn ~ PaperlessBilling, family = binomial(link = "logit"),
    data = training)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -1.65823    0.06062  -27.36  <2e-16 ***
PaperlessBillingyes  0.99392    0.07219   13.77  <2e-16 ***
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5702.8  on 4923  degrees of freedom
Residual deviance: 5496.5  on 4922  degrees of freedom
AIC: 5500.5

Number of Fisher Scoring iterations: 4

> # Evaluate model with testing data
> fitted.resultsP <- predict(LogModelP, newdata = testing, type = "response")
> fitted.resultsP <- ifelse(fitted.resultsP > 0.5, 1, 0)
> misclasificErrorP <- mean(fitted.resultsP != testing$Churn)
> # Display a confusion matrix using the table() function
> confusionmatrixP <- table(Predicted = fitted.resultsP, Actual = testing$Churn)
> confusionmatrixP
      Actual
Predicted 0    1
      0 1548  560
> # Accuracy
> accuracyP <- 1 - misclasificErrorP
> accuracyP
[1] 0.7343454
```

Note. Logistic regression output for the PaperlessBilling-only model, including coefficient estimates, confusion matrix, and model accuracy.

Figure 5

Logistic Regression Model Using tenure_group

```

> #####
> # Create and evaluate logistic model using tenure_group
> LogModelT <- glm(churn ~ tenure_group, family = binomial(link = "logit"), data = training)
> summary(LogModelT)

call:
glm(formula = churn ~ tenure_group, family = binomial(link = "logit"),
    data = training)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)      -2.5235     0.1225  -20.603  < 2e-16 ***
tenure_group0-12 Month    2.4073     0.1327   18.135  < 2e-16 ***
tenure_group12-24 Month    1.6348     0.1478   11.059  < 2e-16 ***
tenure_group24-48 Month    1.1534     0.1428    8.079 6.55e-16 ***
tenure_group48-60 Month    0.6639     0.1738    3.820 0.000134 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5702.8  on 4923  degrees of freedom
Residual deviance: 5091.3  on 4919  degrees of freedom
AIC: 5101.3

Number of Fisher Scoring iterations: 5

> # Evaluate model with testing data
> fitted.resultsT <- predict(LogModelT, newdata = testing, type = "response")
> fitted.resultsT <- ifelse(fitted.resultsT > 0.5, 1, 0)
> misclasificErrorT <- mean(fitted.resultsT != testing$churn)
> # Display a confusion matrix using the table() function
> confusionmatrixT <- table(Predicted = fitted.resultsT, Actual = testing$churn)
> confusionmatrixT
      Actual
Predicted  0    1
      0 1548  560
> # Accuracy
> accuracyT <- 1 - misclasificErrorT
> accuracyT
[1] 0.7343454

```

Note. Logistic regression output for the tenure_group-only model, including coefficient estimates, confusion matrix, and model accuracy.

When the three significant predictors were evaluated collectively in Figure 6, the confusion matrix changed noticeably from the individual predictor models. Unlike the separate models, which classified every case as non-churn, the collective model produced both churn and non-churn predictions. At the 0.5 threshold, the model identified true negatives, false negatives, false positives, and true positives, showing a more balanced classification pattern. It correctly identified 1,398 non-churn customers and 218 churn cases, while 150 non-churn customers and 342 churn cases were misclassified. Although the collective model accuracy was 76.7%, which was slightly lower than the full model in Figure 2, it performed better than the individual predictor models because it no longer defaulted entirely to the majority class. For churn prediction, the collective model was stronger than using a single predictor because it captured the combined influence of contract type, billing behavior, and customer tenure.

Figure 6

Logistic Regression Model Using the Three Most Significant Predictors Collectively

```
> #####
> # Evaluate the three significant predictors collectively
> LogModelCollectively <- glm(Churn ~ Contract + PaperlessBilling + tenure_group,
+                             family = binomial(link = "logit"),
+                             data = training)
> summary(LogModelCollectively)

Call:
glm(formula = Churn ~ Contract + PaperlessBilling + tenure_group,
    family = binomial(link = "logit"), data = training)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   -1.42613    0.17422  -8.186 2.71e-16 ***
ContractOne year  -1.29047    0.11772 -10.962 < 2e-16 ***
ContractTwo year  -2.57656    0.19936 -12.924 < 2e-16 ***
PaperlessBillingYes  0.90380    0.08032  11.253 < 2e-16 ***
tenure_group0-12 Month  0.87188    0.16657   5.234 1.66e-07 ***
tenure_group12-24 Month  0.25492    0.17656   1.444   0.149
tenure_group24-48 Month  0.02213    0.16771   0.132   0.895
tenure_group48-60 Month -0.13087    0.19363  -0.676   0.499
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5702.8  on 4923  degrees of freedom
Residual deviance: 4549.3  on 4916  degrees of freedom
AIC: 4565.3

Number of Fisher Scoring iterations: 6

> # Evaluate model with testing data
> fitted.resultsCollectively <- predict(LogModelCollectively, newdata = testing, type = "response")
> fitted.resultsCollectively <- ifelse(fitted.resultsCollectively > 0.5, 1, 0)
> misclasificErrorCollectively <- mean(fitted.resultsCollectively != testing$Churn)
> # Display a confusion matrix using the table() function
> confusionmatrixCollectively <- table(Predicted = fitted.resultsCollectively, Actual = testing$Churn)
> confusionmatrixCollectively
      Actual
Predicted   0    1
      0 1398  342
      1  150  218
> # Accuracy
> accuracyCollectively <- 1 - misclasificErrorCollectively
> accuracyCollectively
[1] 0.7666034
> #Record Time & Date
> Sys.time()
[1] "2026-04-19 18:00:58 MDT"
```

Note. Logistic regression output for the collective model using Contract, PaperlessBilling, and tenure_group, including coefficient estimates, confusion matrix, and model accuracy.

This type of analysis helps a company identify which customer characteristics are most associated with churn so retention efforts can be more targeted. The results suggest that contract type, paperless billing behavior, and customer tenure each relate to churn, but they are more useful when evaluated together because customer decisions are influenced by multiple factors at the same time. From a business perspective, this means newer customers, customers on month-to-month contracts, and customers using paperless billing may represent groups that deserve

closer attention during retention campaigns. These insights could support several practical retention strategies. Customers on month-to-month contracts may benefit from targeted incentives that encourage movement into longer-term agreements, since longer contracts were associated with lower churn. Newer customers should also receive stronger onboarding support during their first year, because short tenure showed the highest churn risk and suggests that early customer experience may influence long-term retention. For billing behavior, a company could automatically enroll customers in paperless billing while allowing an opt-out for mailed statements, which creates a consistent digital communication channels.